# DTT: Diversification-Type Tipping

Unified mathematical simulation report

Exploratory models of synchrony, aggregate disturbances, homeostatic escape, cascades, hysteresis, and asymmetric restoration

Scope: general mathematics and dynamical systems only. No applied economic interpretation is included.

Purpose: collect the prior simulation tests and numerical outputs into one auditable document, with results clearly labelled by model, assumptions, and test size.

Status: preliminary computational prototypes. Results show what follows under the chosen equations and parameter settings; they do not establish a universal law or empirical validation.

Prepared 13 October 2026

# Executive summary

The simulations explore one possible mechanism: a component system becomes more coherent; under a specified shared-disturbance mapping, its aggregate response variance rises; larger disturbances make it more likely for a bistable state variable to cross an unstable separator; optional downstream components may then fail, while a separate asymmetrically coupled subsystem may restore the original state. Every arrow in this chain depends on explicit modelling assumptions.

Strongest numerical pattern: in the latest dynamic shared-shock test, tipping rose with Kuramoto coupling K from 4.75% (19/400) at K=0 to 100% (400/400) at K=3.2. In an independent-component-noise control with 4,000 trials, 13 tipped (0.325%). In the temporal-variability test, instantaneous R(t) yielded different finite-horizon probabilities from matched constant-RMS controls, suggesting timing/heterogeneity matters within this model.

Critical qualification: synchrony does not universally amplify aggregate disturbances. That effect follows here because component response is orientation-dependent under a shared disturbance. Independent disturbances with zero cross-covariance do not have the same amplification pathway.

Validation still outstanding: time-step convergence, broader parameter sweeps, multiple random seeds/ensembles for oscillator metrics, independent code review, and empirical calibration. Some prior explorations used distinct equations and parameter choices; their outputs are not directly interchangeable.

# Contents and test map


| Section | Topic |
|---|---|
| A | Mathematical core: component-to-aggregate derivation |
| B | Synchronization and effective component count |
| C | Noise-induced escape from homeostasis |
| D | Dynamic shared-shock test and independent-noise control |
| E | Temporal-variability controls |
| F | Stress-threshold event test |
| G | Conditional downstream cascade tests |
| H | Asymmetric coherent restoration |
| I | Hysteresis and alternative exploratory models |
| J | Caveats, consistency checks, and next tests |

# A. Mathematical core: component-to-aggregate derivation

The component model uses N phase oscillators following noisy Kuramoto dynamics:

d□[?0xd□[0m□[?0x □[0m□[?0x□[0m□[?0xθ□[0mᵢ = [ωᵢ + K Im(Z e⁻ⁱθᵢ)] dt + √(2D) dWᵢ, Z = (1/N) Σⱼ eⁱθⱼ, R = |Z|.

Here K is coupling, D is oscillator diffusion, ωᵢ are natural frequencies, and R is the Kuramoto order parameter: R near zero represents phase dispersion; R near one represents strong phase alignment.

Map each component phase to an orientation vector vᵢ=(cos θᵢ, sin θᵢ). Let a shared disturbance point in random direction u(φ)=(cos φ, sin φ), and define component response qᵢ=σ u·vᵢ. The aggregate response is q̄=σ u·v̄, where v̄=(1/N)Σvᵢ and |v̄|=R.

For uniformly random φ, conditional on the oscillator phases: Eφ[q̄]=0 and Varφ(q̄)=σ²R²/2. Thus synchrony raises aggregate response variance in this model through vector alignment under a common directional disturbance.

For random independent phases, E[R²]=1/N. The variance amplification relative to that incoherent baseline is A=N E[R²], while N_eff=1/E[R²] is a variance-equivalent effective count. These are model-derived metrics, not literal universal counts of subsystems.

Assumption checkpoint: if components instead receive independent disturbances with no cross-covariance, phase alignment alone does not create the same shared-disturbance amplification. The shared shock and orientation-dependent response are essential model ingredients.

# B. Synchronization and diversification proxy

Oscillator parameters in the component-derived run: N=31; 40 independent ensembles; natural frequencies drawn from Normal(0,1); initial phases uniform on [-π,π]; oscillator diffusion D=0.015; coupling K swept from 0 to 3.2. Parameters are illustrative, not fitted to observations.

| K | mean R2 | theoretical var q over sigma2 | MC var q over sigma2 | theoretical amplification | Neff from aggregate variance |
|---|---|---|---|---|---|
| 0.000 | 0.0318 | 0.01591 | 0.01590 | 0.99x | 31.42 |
| 0.267 | 0.0398 | 0.01988 | 0.01996 | 1.23x | 25.16 |
| 0.533 | 0.0536 | 0.02679 | 0.02675 | 1.66x | 18.67 |
| 0.800 | 0.0733 | 0.03664 | 0.03650 | 2.27x | 13.65 |
| 1.067 | 0.0976 | 0.04878 | 0.04911 | 3.02x | 10.25 |
| 1.333 | 0.1711 | 0.08555 | 0.08573 | 5.30x | 5.84 |
| 1.600 | 0.3029 | 0.15144 | 0.15202 | 9.39x | 3.30 |
| 1.867 | 0.4403 | 0.22015 | 0.22100 | 13.65x | 2.27 |
| 2.133 | 0.6182 | 0.30912 | 0.30839 | 19.17x | 1.62 |
| 2.400 | 0.7532 | 0.37659 | 0.37808 | 23.35x | 1.33 |
| 2.667 | 0.8168 | 0.40838 | 0.40609 | 25.32x | 1.22 |
| 2.933 | 0.8625 | 0.43126 | 0.43021 | 26.74x | 1.16 |
| 3.200 | 0.8911 | 0.44554 | 0.44581 | 27.62x | 1.12 |

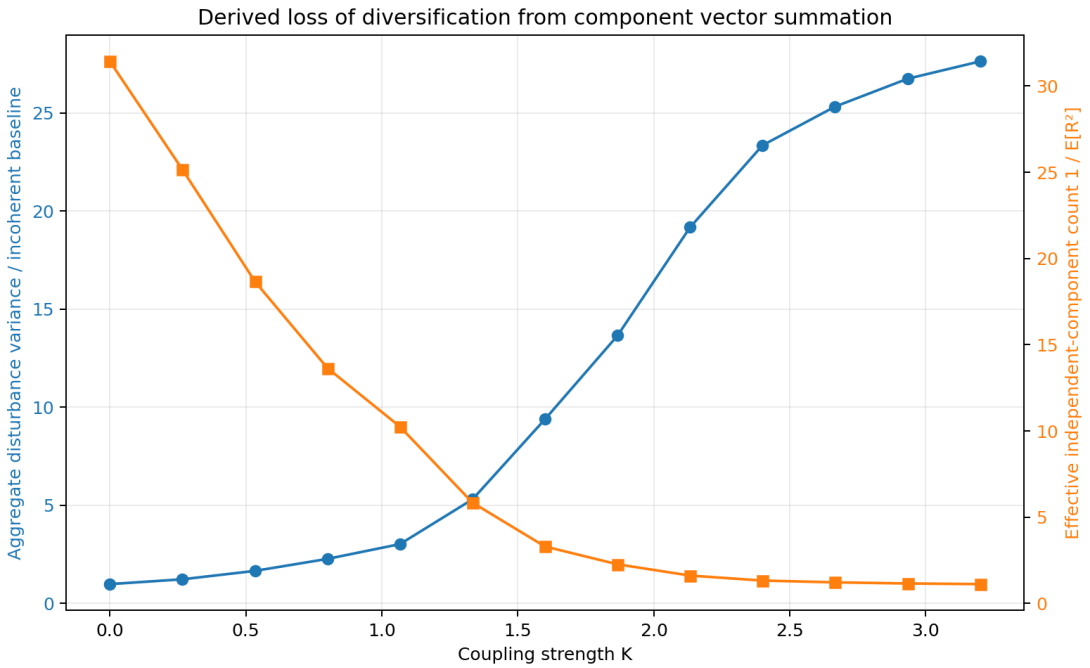



Figure B1. Component-derived variance amplification and effective count versus coupling; values follow the specified phase-response mapping.

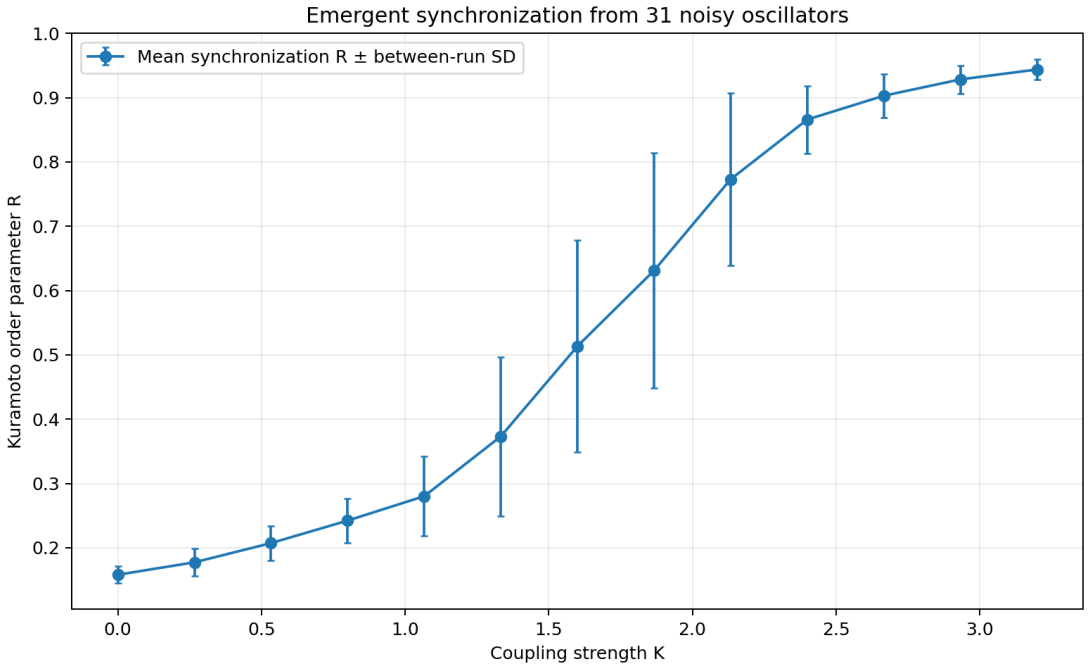



Figure B2. Synchronization metrics in the component-derived model.

The theoretical and Monte Carlo values for the directional-response variance broadly agree across K. That verifies the algebra-to-simulation bridge for this generated model. It does not validate whether this phase-to-response mapping describes a real system.

# C. Noise-induced escape from the original homeostatic state

A bistable scalar normal form was used: $dx=(\mu+x-x^3)dt + noise \cdot dW$, with $\mu=0.20$. The deterministic equilibria are approximately $x_l=-0.878885$ (stable), $x_u=-0.209149$ (unstable separator), and $x_r=1.088034$ (stable). Initial state is $x_l$; a tipping event is first passage to $x \geq 0$ within a finite horizon of 40 time units. The deterministic drift is held fixed; the synchronization-derived noise scale changes. Hence this is noise-induced escape, not a deterministic bifurcation.

| K | mean R | mean R2 | N eff 1 over mean R2 | crossings | trials | crossing probability | wilson 95 low | wilson 95 high | median first passage time conditional |
|---|---|---|---|---|---|---|---|---|---|
| 0.000 | 0.158 | 0.032 | 31.42 | 2 | 500 | 0.4% | 0.1% | 1.4% | 9.14 |
| 0.800 | 0.242 | 0.073 | 13.65 | 118 | 500 | 23.6% | 20.1% | 27.5% | 26.54 |
| 1.333 | 0.373 | 0.171 | 5.84 | 422 | 500 | 84.4% | 81.0% | 87.3% | 12.51 |
| 1.867 | 0.631 | 0.440 | 2.27 | 500 | 500 | 100.0% | 99.2% | 100.0% | 4.67 |
| 3.200 | 0.944 | 0.891 | 1.12 | 500 | 500 | 100.0% | 99.2% | 100.0% | 2.29 |

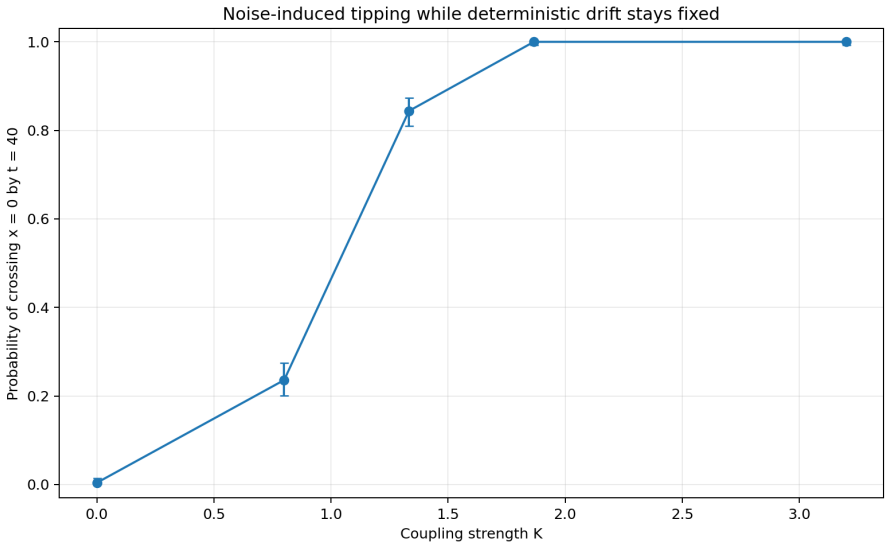



Figure C2. Finite-horizon tipping probability under the component-derived, averaged-noise model.

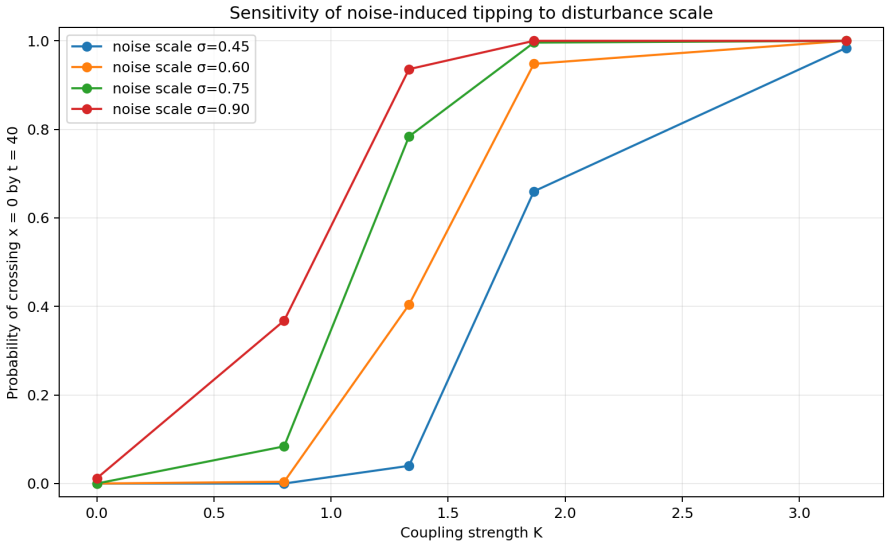



Figure C3. Sensitivity of tipping probability to noise scale and coupling.

Reported base sweep (500 trials per K): K=0: 2/500 (0.4%); K=0.8: 118/500 (23.6%); K=1.333: 422/500 (84.4%); K=1.867: 500/500 (100%); K=3.2: 500/500 (100%). Median conditional first-passage times were 12.505, 4.670, and 2.285 time units for K=1.333, 1.867, and 3.2 respectively. These are a separate simulation run from the later dynamic-shock experiment below.

# D. Dynamic shared-shock test versus independent-noise control

Here oscillator trajectories and the stress state were integrated simultaneously, with N=31, $\omega_i \sim$Normal(0,1), initial phases uniform on [-π,π], oscillator diffusion D=0.02, μ=0.2, σ=0.82, Euler-Maruyama step dt=0.01, horizon T=40, and initial x=-0.878885. Shared-disturbance case used state noise σR(t); independent-component control used σ/√N, independent of K. Shared-shock runs used 400 trials per K; independent control used 4,000 trials.

| K | trials | crossings | tipping probability | wilson95 low | wilson95 high | median tip time among tippers | share control |
|---|---|---|---|---|---|---|---|
| 0.0 | 400 | 19 | 4.75% | 3.06% | 7.30% | 27.71 | shared disturbance; instantaneous amplitude sigma*R(t) |
| 0.8 | 400 | 136 | 34.00% | 29.53% | 38.77% | 21.54 | shared disturbance; instantaneous amplitude sigma*R(t) |
| 1.3333333333 | 400 | 296 | 74.00% | 69.49% | 78.06% | 14.64 | shared disturbance; instantaneous amplitude sigma*R(t) |
| 1.8666666667 | 400 | 388 | 97.00% | 94.83% | 98.28% | 7.67 | shared disturbance; instantaneous amplitude sigma*R(t) |
| 3.2 | 400 | 400 | 100.00% | 99.05% | 100.00% | 3.82 | shared disturbance; instantaneous amplitude sigma*R(t) |
| K-invariant | 4000 | 13 | 0.33% | 0.19% | 0.56% | 24.72 | independent component disturbances; aggregate amplitude sigma/sqrt(N) |

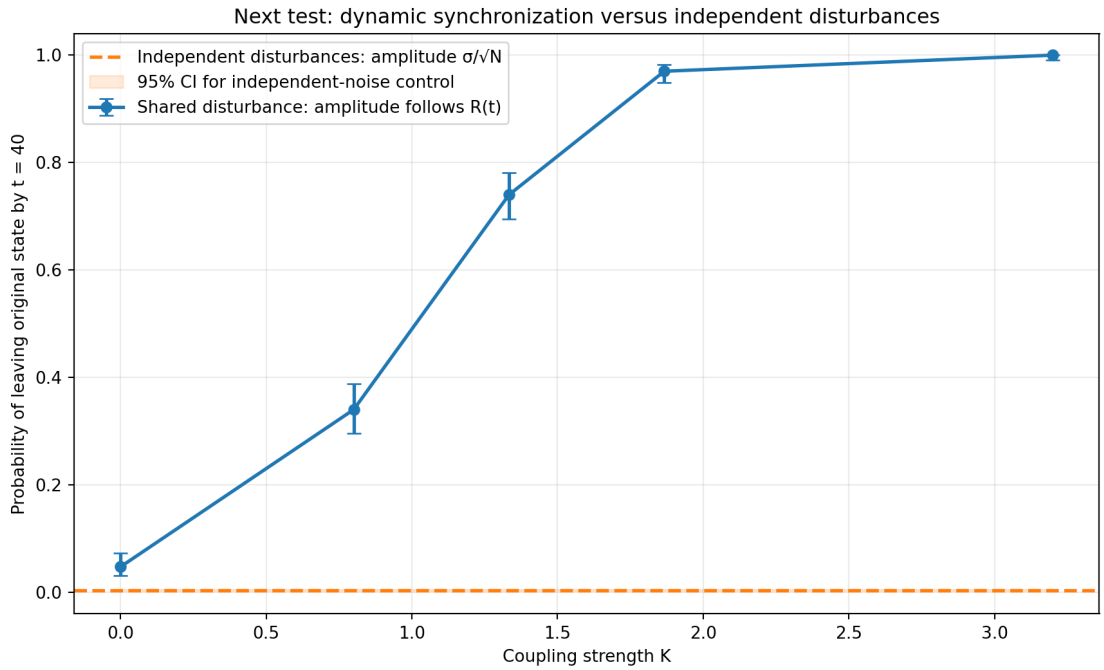



Figure D1. Dynamic shared-shock tipping results compared with an independent-component-noise control.

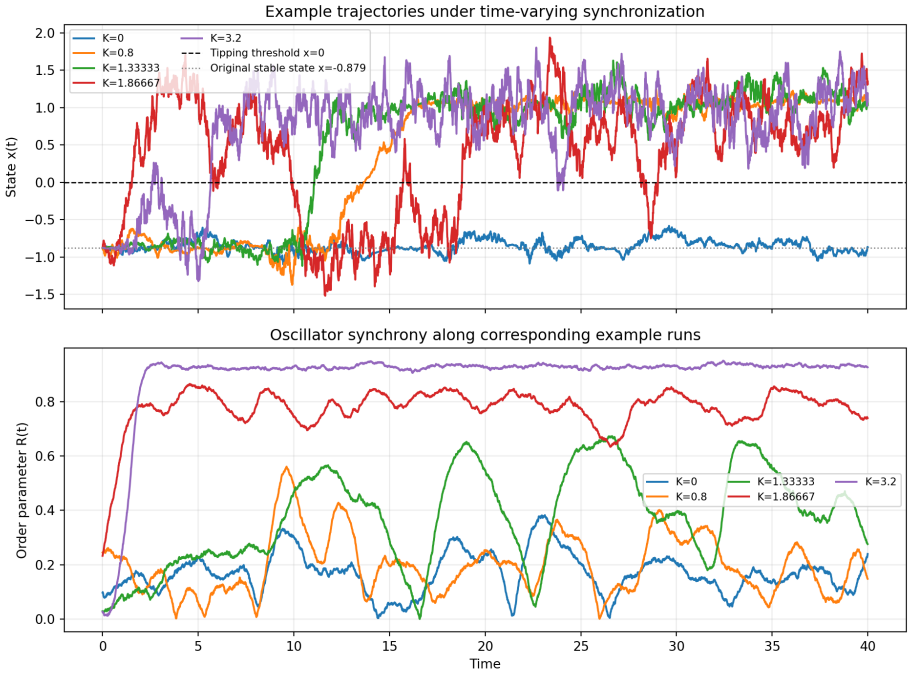



Figure D2. Example state trajectories from the dynamic test; individual random paths can vary.

Outcome: the shared-shock run reported tipping rates of 4.75%, 34%, 74%, 97%, and 100% at K=0, 0.8, 1.333, 1.867, and 3.2. The K-invariant independent-noise control tipped in 13/4,000 trials (0.325%, Wilson 95% CI 0.190%-0.555%). The comparison is consistent with the proposed mechanism within this model, but model assumptions and numerical convergence still need independent checks.

# E. Temporal-variability controls

This follow-up isolates whether using instantaneous R(t) matters beyond a constant RMS disturbance. It used the same core parameters (N=31, D=0.02, μ=0.2, σ=0.82, dt=0.01, T=40), with 800 trials per K for each of three paired formulations: (1) instantaneous amplitude σR(t); (2) each trial's constant amplitude $\sigma\sqrt{\text{mean}_t R(t)^2}$; and (3) K-specific ensemble RMS amplitude $\sigma\sqrt{\text{mean}_{(\text{trials},t)} R(t)^2}$. Matched Brownian increments were used across the three formulations within each K/trial. Independent-noise control used 4,000 trials.

| K | noise condition | trials | tipping count | tipping probability | ci95 low | ci95 high | median tipping time |
|---|---|---|---|---|---|---|---|
| 0.000 | Instantaneous R(t) | 800 | 22 | 2.75% | 1.82% | 4.13% | 23.29 |
| 0.000 | Per-realization constant RMS | 800 | 5 | 0.62% | 0.27% | 1.45% | 18.26 |
| 0.000 | K-specific ensemble-mean RMS | 800 | 4 | 0.50% | 0.19% | 1.28% | 15.12 |
| 0.800 | Instantaneous R(t) | 800 | 263 | 32.88% | 29.71% | 36.21% | 16.19 |
| 0.800 | Per-realization constant RMS | 800 | 161 | 20.12% | 17.49% | 23.04% | 16.86 |
| 0.800 | K-specific ensemble-mean RMS | 800 | 160 | 20.00% | 17.37% | 22.91% | 19.28 |
| 1.333 | Instantaneous R(t) | 800 | 590 | 73.75% | 70.59% | 76.68% | 14.13 |
| 1.333 | Per-realization constant RMS | 800 | 531 | 66.38% | 63.03% | 69.56% | 12.79 |
| 1.333 | K-specific ensemble-mean RMS | 800 | 663 | 82.88% | 80.11% | 85.33% | 13.76 |
| 1.867 | Instantaneous R(t) | 800 | 773 | 96.62% | 95.13% | 97.67% | 7.85 |
| 1.867 | Per-realization constant RMS | 800 | 767 | 95.88% | 94.26% | 97.05% | 5.28 |
| 1.867 | K-specific ensemble-mean RMS | 800 | 797 | 99.62% | 98.90% | 99.87% | 5.29 |
| 3.200 | Instantaneous R(t) | 800 | 800 | 100.00% | 99.52% | 100.00% | 3.82 |

| K | noise condition | trials | tipping count | tipping probability | ci95 low | ci95 high | median tipping time |
|---|---|---|---|---|---|---|---|
| 3.200 | Per-realization constant RMS | 800 | 800 | 100.00% | 99.52% | 100.00% | 2.24 |
| 3.200 | K-specific ensemble-mean RMS | 800 | 800 | 100.00% | 99.52% | 100.00% | 2.21 |

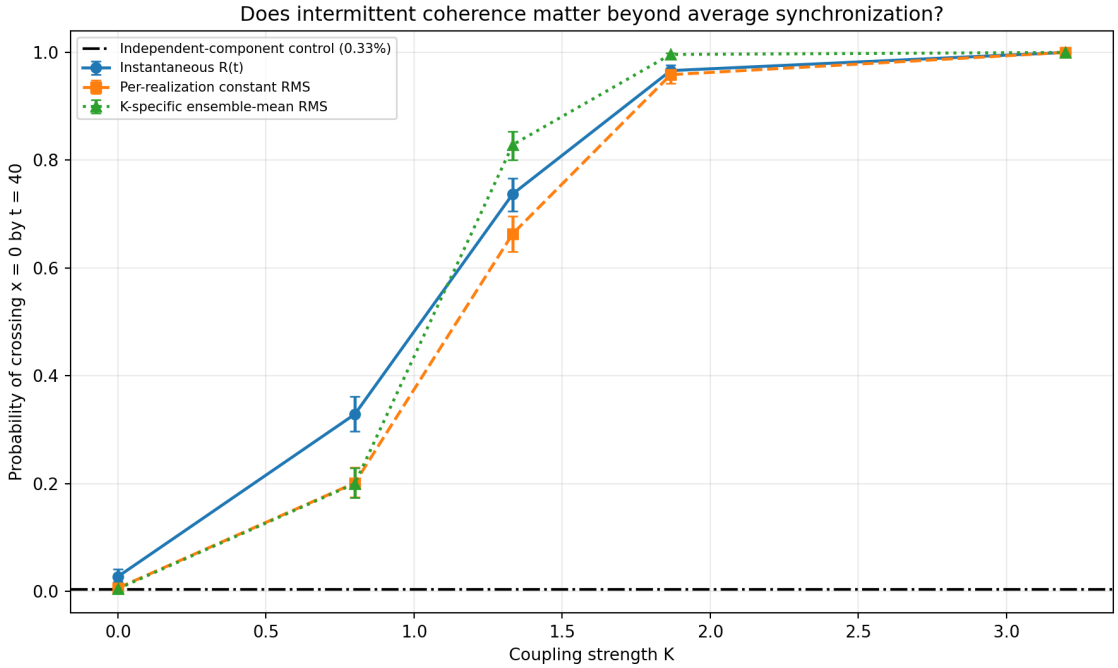



Figure E1. Comparison of instantaneous R(t) noise and constant-RMS control formulations.

At K=0.8, instantaneous R(t) gave 32.875% tipping (263/800) versus 20.125% (161/800) with per-realization constant RMS and 20.0% (160/800) with K-specific ensemble RMS. At K=3.2 all three formulations tipped in 800/800 trials. At K=0, event rates were low but different: 2.75% instantaneous versus 0.625% and 0.5% in the two constant controls. This points to an effect of time variation in this finite-horizon simulation; it does not establish a universal temporal-burst theorem.

Run reconciliation: the dynamic test's 400-trial result at K=0.8 was 34%; the later 800-trial temporal-variability run reports 32.875%. These are distinct Monte Carlo runs and should not be merged as though they were one sample. The same applies to the differing reported K=0 tipping estimates.

# F. Separate stress-threshold event test

An earlier abstract prototype used a discrete-time stress state $x[t+1]=a\,x[t]+\sigma\sqrt{(V/N)}\,\varepsilon[t]$, with $a=0.90$, $\sigma=0.25$, and an event defined as $|x|\geq 0.80$ for two consecutive steps. N=31. This test measures thresholded stress events, not escape across a bistable equilibrium.

| scenario | K | R | V | N eff | innovation sd | stationary sd theory | threshold event probability | median event time steps | trials |
|---|---|---|---|---|---|---|---|---|---|
| Asynchronous baseline | 0 | 0.1611 | 1 | 31 | 0.0449 | 0.103 | 0 | - | 600 |
| Intermediate coupling | 1.5 | 0.5876 | 10.7 | 2.896 | 0.1469 | 0.337 | 0.935 | 202 | 600 |
| Strong synchronization | 2.5 | 0.9367 | 27.2 | 1.14 | 0.2342 | 0.5372 | 1 | 34 | 600 |

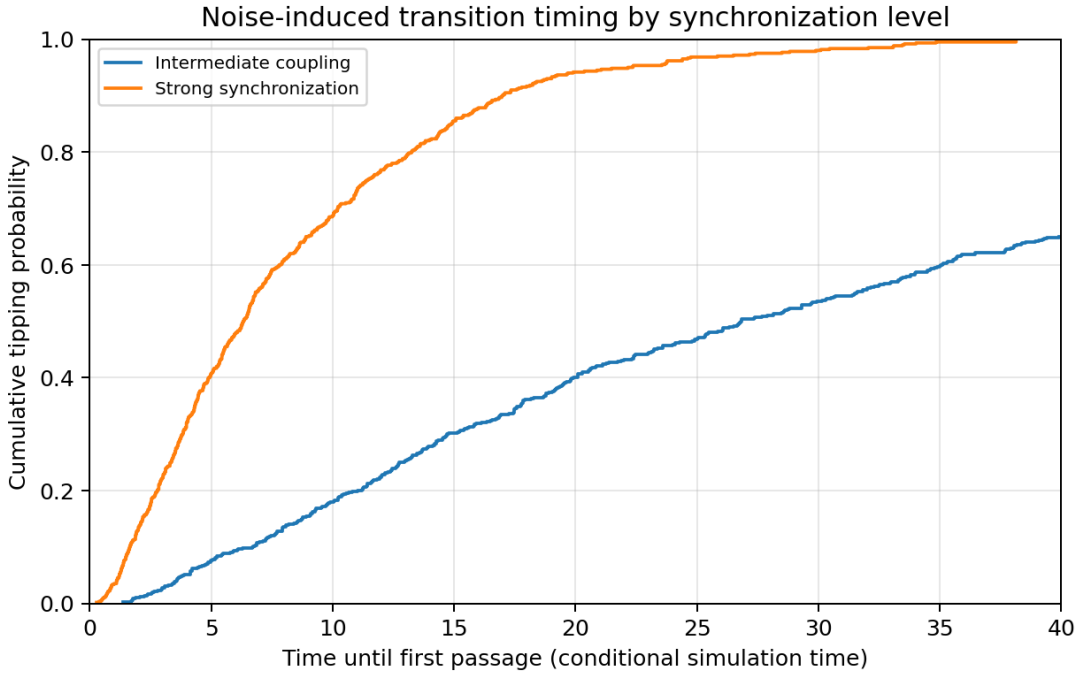



Figure F1. Timing plot from the early abstract event/cascade prototype.

Reported over 600 trials and 800 steps: asynchronous baseline K=0 had 0 threshold events; intermediate K=1.5 had 561/600 events (93.5%), median event time 202 steps; strong synchronization K=2.5 had 600/600 events (100%), median time 34 steps. This is a different operational event definition and model from the cubic tipping model.

# G. Conditional downstream-cascade tests

Two exploratory cascade modules were tested separately from the synchrony/noise mechanism. In the first, component A adapts slowly; components B-D have finite error tolerances and fail only after stress has propagated through a chosen coupling. In a second Monte Carlo sensitivity test, five dependent units had random reserves and could fail if transferred stress remained above reserve for a grace period. These modules illustrate how a primary transition could be followed by cascading failure; they are not mathematically implied by the oscillator model.

| Component | Adaptation time constant | Error tolerance | Failure time without cascade | Failure time with cascade | Failed only when cascade coupling enabled |
|---|---|---|---|---|---|
| A (slow) | 2 | 0.5 | 74.16 | 74.16 | False |
| B | 1.2 | 0.5 | - | 74.42 | True |
| C | 1.2 | 0.5 | - | 76.01 | True |
| D | 1.2 | 0.5 | - | 77.6 | True |

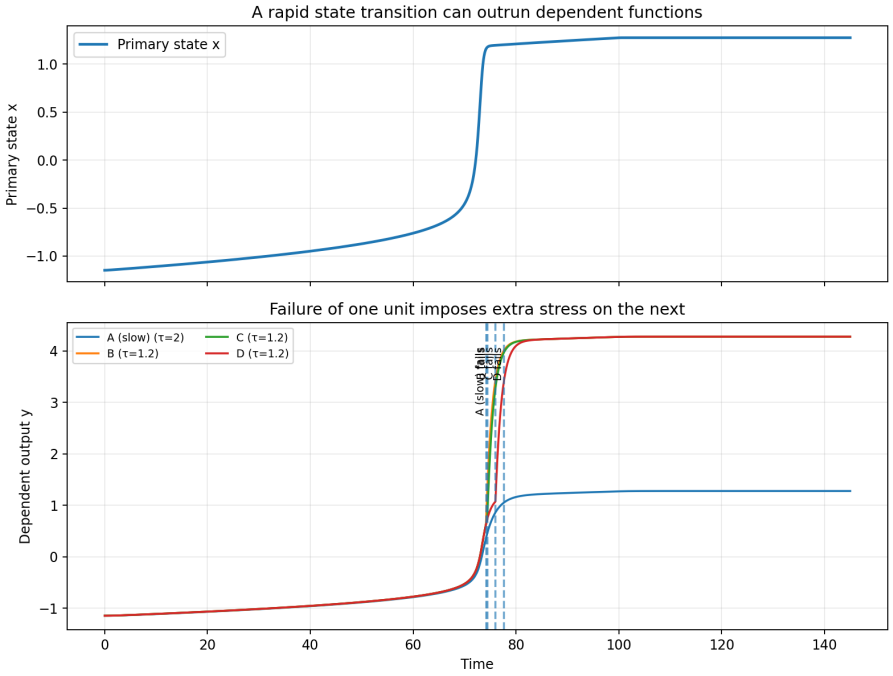



Figure G1. Illustrative delayed downstream cascade trajectories.

| primary event severity | dependent units | trials | reserve distribution | base stress coefficient | transfer per failed unit | grace steps | mean units failed | probability any dependent failure |
|---|---|---|---|---|---|---|---|---|
| 0.85 | 5 | 1000 | Uniform(0.70, 1.15) | 0.65 | 0.25 | 2 | 0 | 0 |
| 1 | 5 | 1000 | Uniform(0.70, 1.15) | 0.65 | 0.25 | 2 | 0 | 0 |
| 1.15 | 5 | 1000 | Uniform(0.70, 1.15) | 0.65 | 0.25 | 2 | 2.096 | 0.428 |
| 1.3 | 5 | 1000 | Uniform(0.70, 1.15) | 0.65 | 0.25 | 2 | 4.26 | 0.852 |

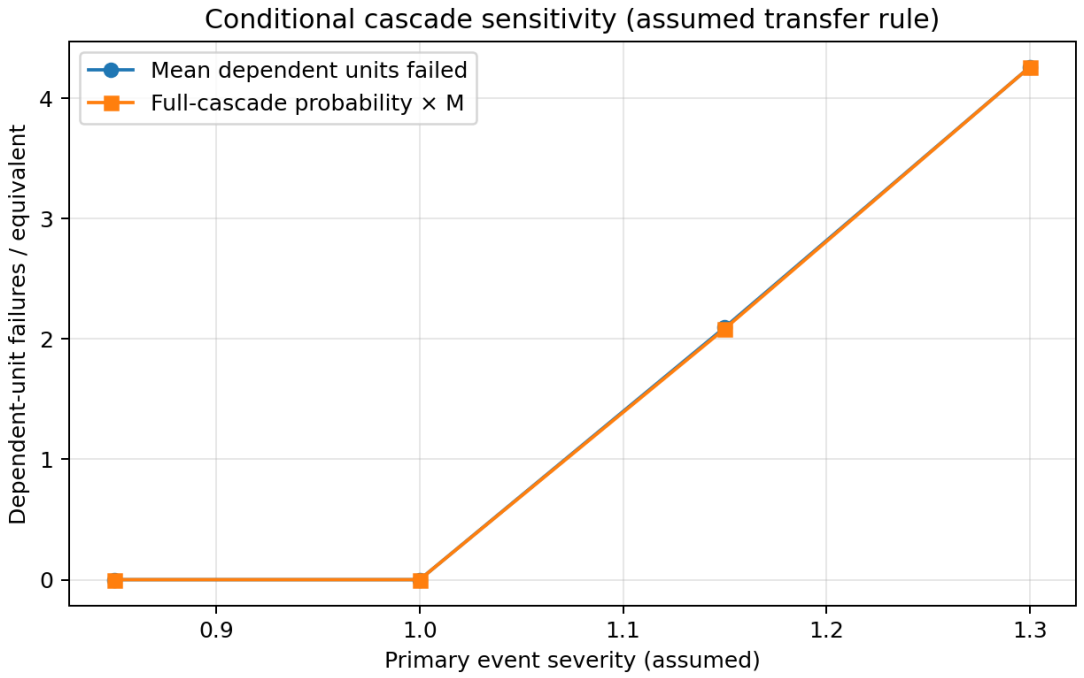



Figure G2. Cascade sensitivity in the five-dependent-unit prototype.

In the five-unit sensitivity table (1,000 trials per severity), at severities 0.85 and 1.00 no dependent failures were reported. At 1.15, probability of any dependent failure was 42.8% and full cascade 41.7%; at 1.30, both were 85.2%. Since the model's transfer/stress rules are stipulated, this demonstrates conditional cascade behavior, not a general law.

# H. Asymmetric coherent restoration

A separate toy model tested the conceptual case in which a larger subsystem remains macroscopically near its reference state while a coherence-related coupling corrects a smaller subsystem. This implements restoration through an external asymmetric coupling, not self-restoration by the smaller state.

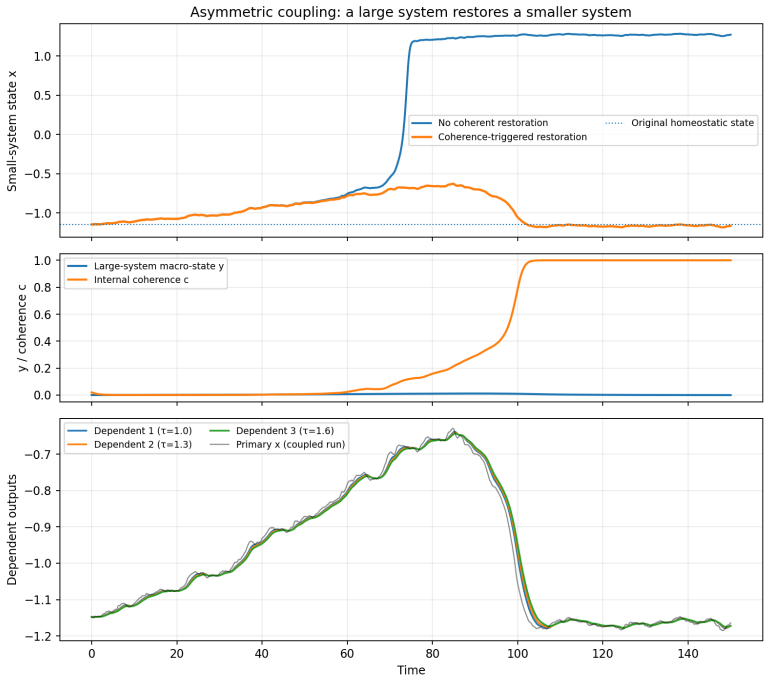



Figure H1. Asymmetric coherent-restoration prototype.

| Metric | Value |
|---|---|
| Original homeostatic state x0 | -1.148 |
| Minimum x with restoration | -1.185 |
| Maximum x with restoration | -0.6292 |
| Mean x over final 10 time units (with restoration) | -1.166 |
| Mean x over final 10 time units (no restoration) | 1.271 |
| Final coherence c | 1 |
| Maximum large-system macro-state \|y\| | 0.01136 |
| Maximum x departure from original state | 0.5185 |
| Restoration error at final time | 0.01624 |
| Dependent 1 failure time | - |
| Dependent 2 failure time | - |
| Dependent 3 failure time | - |

Reported summary: original homeostatic state $x_0$=-1.147677; with restoration, mean x over the final 10 time units was -1.165784 versus +1.270615 without restoration; final coherence c=1; maximum macro-state deviation |y|=0.011356; final restoration error 0.016236. These figures show the desired behavior under chosen equations and coupling, not robustness over a broad parameter domain.

# I. Hysteresis and other exploratory aggregation models

Several exploratory models checked related but distinct ideas: weighted aggregation/cancellation, a self-reinforcing two-state model, parameter sweeps for hysteresis, and response competition. These are included as exploratory tests, not as evidence that all mechanisms are the same. In particular, some uncorrected prototypes did not exhibit hysteresis and should not be presented as successful positive tests.

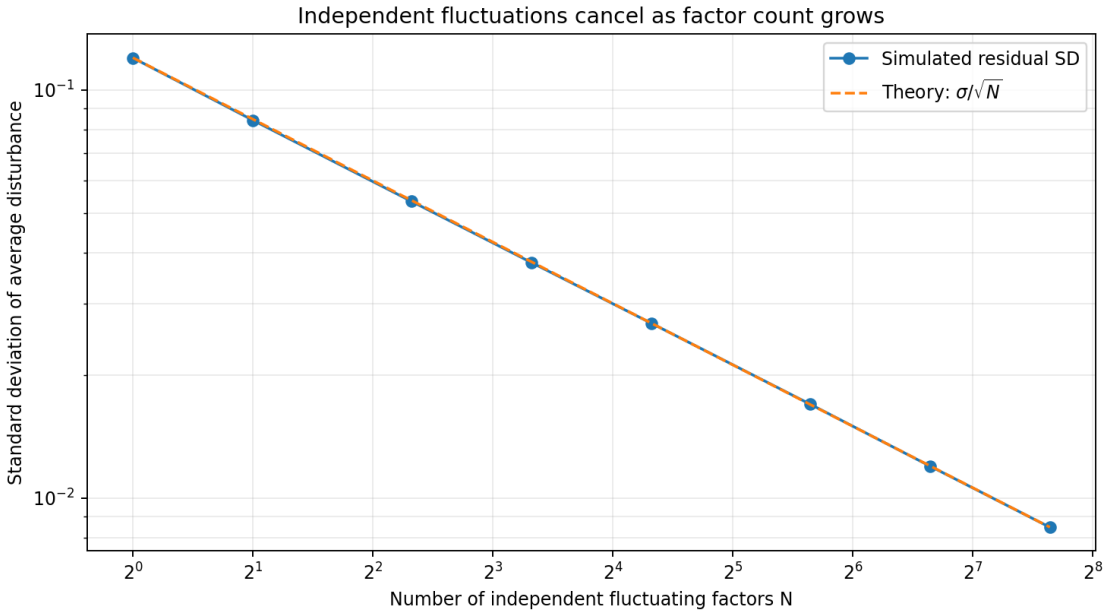



Figure I1. Factor cancellation/aggregation example.

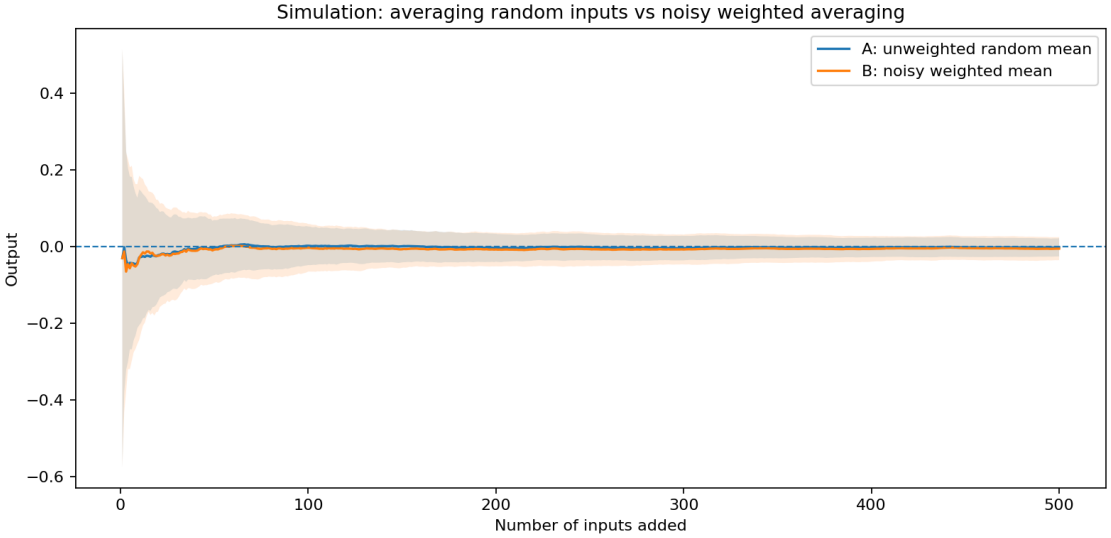



Figure I2. Weighted aggregation example.

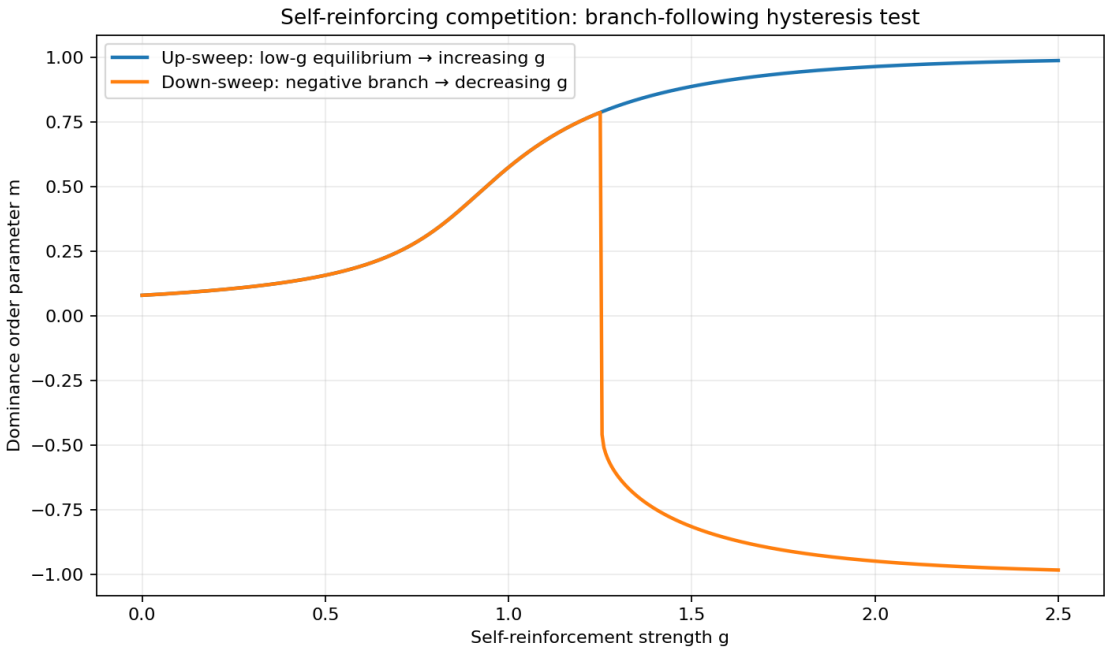



Figure I3. Corrected self-reinforcing hysteresis prototype.

| Metric | Value |
|---|---|
| Baseline bias h | 0.08 |
| Largest up-sweep jump near g | 0.915 |
| Largest down-sweep jump near g | 1.25 |
| Maximum branch gap | 1.971 |
| Approximate hysteresis-loop area | 2.273 |
| Up-sweep state at g=2.5 | 0.9879 |
| Down-sweep state at g=2.5 | -0.9829 |
| Down-sweep state at g=0 | 0.07983 |

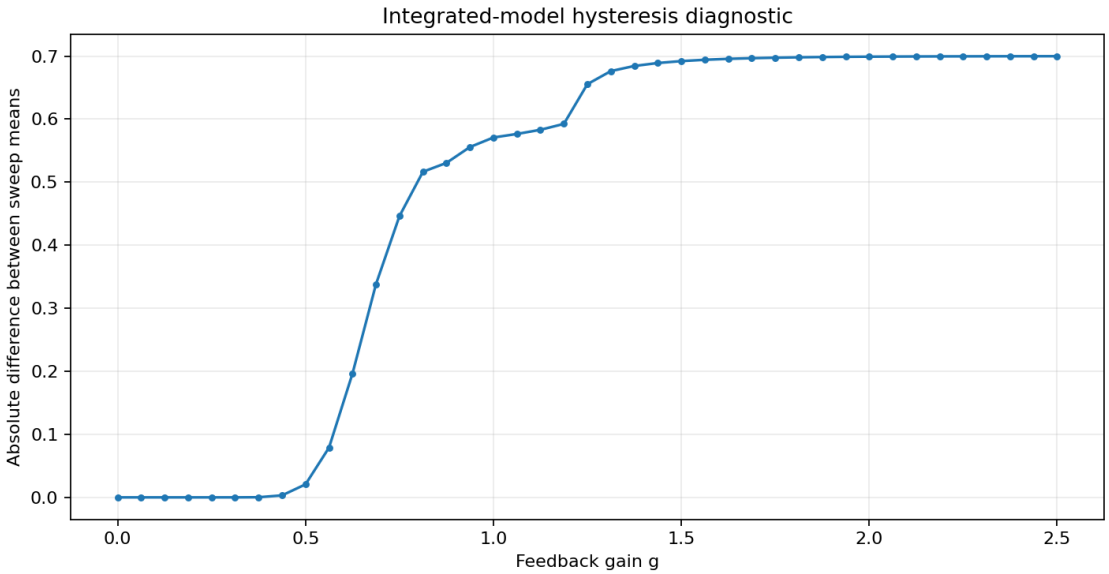



Figure I4. Integrated-model up/down sweep comparison.

| Metric | Value |
|---|---|
| Max \|state_up - state_down\| | 2.977e-06 |

| Metric | Value |
|---|---|
| Mean \|state_up - state_down\| | 9.579e-08 |
| Max \|output_up - output_down\| | 3.086e-06 |
| Up-sweep half-level crossing g | 0.9 |
| Down-sweep half-level crossing g | 0.9 |
| State at g=0, up-sweep | 0 |
| State at g=2, up-sweep | 0.9998 |

Hysteresis reconciliation: one corrected self-reinforcing model reported a maximum branch gap of 1.970796 and approximate loop area 2.273208. A separate “hysteresis test” reported maximum $|state_up-state_down| \approx 2.98\times10^{-6}$, effectively no hysteresis. Those are different model formulations; the near-zero-gap result must remain visible as a negative/control result.

# J. Overall evidence audit and conclusions

## What the tests support

1. The component-vector derivation yields $Var(\bar{q}|phases)=\sigma^2R^2/2$ for the specified shared directional shock. The Monte Carlo variance check tracks the analytical expression closely.

2. Under the selected cubic bistable state equation, increasing the synchronization-linked noise scale increases finite-horizon escape probability across the reported sweeps.

3. In the dynamic shared-shock model, tipping was much more frequent than in the K-invariant independent-noise control. Constant-RMS controls further suggest that time variation of R(t) can affect escape probability in this parameter regime.

4. Conditional cascade and asymmetric restoration behaviors can be produced by additional coupling rules. They require their own validation and cannot be deduced from the synchrony model alone.

## What the tests do not establish

1. They do not establish a universal mechanism or a named scientific law. All results depend on chosen equations, parameter values, event thresholds, and couplings.

2. They do not show that synchronization always reduces “diversity” in a general system; here N_eff is a specific variance-equivalent measure under a specified response model.

3. They do not establish numerical robustness yet: Euler-Maruyama time-step convergence and broader parameter sweeps remain outstanding. Monte Carlo uncertainty intervals account for finite trial count, not model uncertainty or implementation errors.

4. They do not establish empirical validity. No external data have calibrated the phase-response mapping, noise scale, cascade rules, or restoration coupling.

## Recommended next simulation sequence

• Numerical convergence: repeat key dynamic tests at dt=0.02, 0.01, 0.005, and 0.0025 with common random-number design where possible.

• Parameter robustness: vary oscillator diffusion D, frequency dispersion, N, μ, σ, and the tipping horizon, and report confidence intervals for each condition.

• Mechanism ablation: keep the same R(t) traces but permute their time order; compare instantaneous R(t), constant RMS, shuffled R(t), and independent noise.

• Covariance sensitivity: vary common-shock fraction from 0 to 1 continuously rather than comparing only fully shared and independent disturbances.

• Independent reproduction: have the code and result tables reviewed and regenerate all plots from a clean script before claiming numerical reproducibility.

• For cascades/restoration, map parameter regions where the behavior persists rather than showing only one illustrative parameterization.

# Appendix: key equations and operational definitions

Order parameter: $Z=N^{-1}\Sigma e^{i}\theta_{i}$; $R=|Z|$.
Directional component response: $q_{i}=\sigma u(\varphi)\cdot v_{i}$.
Aggregate shared-response variance: $Var_{\varphi}(\bar{q}|\theta)=\sigma^{2}R^{2}/2$.
Variance amplification: $A=N\ E[R^{2}]$.
Variance-equivalent effective count: $N_{eff}=1/E[R^{2}]$.
Bistable stress state: $dx=(\mu+x-x^{3})dt+g(t)dW$; $\mu=0.20$; lower stable $x\approx-0.878885$, unstable separator $x\approx-0.209149$, upper stable $x\approx1.088034$.
Tipping event in cubic tests: first crossing $x\geq0$ within stated horizon.
Dynamic shared-shock model: $g(t)=\sigma R(t)$.
Independent-noise control: $g=\sigma/\sqrt{N}$, fixed independent of K.
Confidence intervals: Wilson 95% intervals are shown where stored by the simulation outputs. These reflect trial-count uncertainty only.